

David Cai

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EDUCATION

University of Texas at Arlington

Bachelor of Science in Computer Science

Arlington, TX

Dec 2027

University of California, Irvine

Full Stack Development Certificate

Irvine, CA

Oct 2018 – Apr 2019

EXPERIENCE

Software Engineer Intern, Technology Internship Program

Jun 2026 – Aug 2026

Capital One

Plano, TX

- Built and shipped an AI summarization pipeline in Python/JupyterHub that ingests live loan application data via internal APIs, reducing context-gathering time from 5 minutes to 2 minutes per case for 100+ auto loan funding users.
- Implemented retrieval-augmented generation (RAG) using a Chroma vector store and OpenAI embeddings to ground model summaries, eliminating critical hallucinations and ensuring output accuracy.
- Spearheaded end-to-end product discovery and user research, running weekly surveys and live feedback demos with end users to iterate on prompt structures and retrieval top-k parameters, achieving an 80% user-rated accuracy score.
- Partnered cross-functionally with PMs, Data Scientists, and UI/UX designers to translate user friction points into prioritized technical requirements, sprint goals, and system trade-offs.
- Accelerated workflow development via AI-assisted engineering and established comprehensive system documentation, architecture diagrams, and testing handoffs to transition the prototype into production.

PROJECTS

GraphT | PostgreSQL, React/TypeScript, Vercel

Jul 2026 – Present

- Built a crowd-sourced skincare review platform using Next.js 16, Neon Postgres, and Vercel Blob to solve consumer credibility gaps by replacing subjective reviews with standardized, data-backed product trials.
- Integrated the Perfect Corp YouCam AI skin analysis API to capture objective, quantitative skin metrics over time, enabling users to evaluate product efficacy through empirical data rather than qualitative claims.
- Developed a computer vision preprocessing pipeline (BlazeFace) with quality gating and lighting/pose normalization to ensure uniform photo captures for accurate AI skin comparisons.
- Built a web-scraping pipeline to aggregate a catalog of 180k+ skincare products and 20k+ ingredients, allowing users to analyze routines, compare skin profiles, and explore targeted formulations.
- Designed a privacy-focused media architecture on Vercel Blob and Neon Postgres, handling secure storage of user trial photos, routine logs, and multi-metric skin analysis records.

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, HTML/CSS, Python, C, Java, SQL

Frameworks: React, Express.js, LangGraph, SQLAlchemy

AI/ML & Data: Retrieval-Augmented Generation (RAG), Anthropic Claude API, OpenAI API, Chroma, pandas, JupyterHub

Developer Tools: Git, GitHub, GitHub Actions, Vercel, Supabase, Google Cloud Platform, VS Code, IntelliJ, Eclipse,

PostgreSQL, MongoDB, RESTful APIs, Node.js

Libraries: Redux, TanStack Query, Recharts, Tailwind CSS, shadcn/ui, Mantine UI, Chakra UI

STUDENT ORGANIZATION

Treasurer - Mobi Byte (Web Development Club)

Jan 2025 – Present

University of Texas at Arlington

Arlington, TX

- Responsible for managing club finances, including taxes, budgeting, funding allocations, purchase approvals, and financial audits.
- Active engagement with overall club strategy, operations, and event planning as part of the leadership team.
- Mentoring new members on development practices and tools, fostering an inclusive learning environment.
- Design and lead technical workshops on web development topics to support member growth and engagement.